



Reg. No. 2149

New Light Institute

The Finest Institute for Medical Entrance Examination

PHYSICS FORMULAE AND IMPORTANT POINTS

BOOKLET

PHYSICS

1. Least count of Screw Gauge

$$= \frac{\text{Value of one part on main scale } (S)}{\text{Number of parts on circular scale } (n)}$$

2. Average Speed, $v_{av} = \frac{\text{Distance travelled } (\Delta s)}{\text{Time taken } (\Delta t)}$

$$\text{Average Velocity, } \vec{v}_{av} = \frac{\text{Displacement } (\Delta \vec{r})}{\text{Time taken } (\Delta t)}$$

3. Average Acceleration,

$$\vec{a}_{av} = \frac{\text{Change in Velocity } (\Delta \vec{v})}{\text{Time taken } (\Delta t)}$$

4. Instantaneous Speed, $v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$

$$\text{Instantaneous Velocity, } \vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

5. When a particle covers one-third distance at speed v_1 , next one-third at speed v_2 and last one-third at speed v_3 , then

$$v_{av} = \frac{3v_1v_2v_3}{v_1v_2 + v_2v_3 + v_3v_1}$$

6. For uniformly accelerated motion ($\vec{a} = \text{constant}$) equations of motion are as under

$$\vec{v} = \vec{u} + \vec{a}t \quad \dots(i)$$

$$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2 \quad \dots(ii)$$

$$\vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + 2\vec{a} \cdot \vec{s} \quad \dots(iii)$$

Where

$\vec{u}$ = initial velocity, $\vec{v}$ = velocity at time t

$\vec{s}$ = displacement of particle at time t

7. When particle is not uniformly accelerated i.e., acceleration of particle is not constant or acceleration is a function of time then the following relations hold for 1-dimensional motion.

$$(i) v = \frac{ds}{dt} \quad (ii) a = \frac{dv}{dt} = v \frac{dv}{ds}$$

$$(iii) ds = v dt \quad (iv) dv = a dt \quad \text{or } v dv = a ds$$

8. For straight line motion upward and downward motion under the effect of gravity, then

$$v = u \pm gt \quad h = ut \pm \frac{1}{2}gt^2$$

$v^2 = u^2 \pm 2gh$ where g = acceleration due to gravity and take +ve to downward and -ve to upward motion,

9. For two particles having displacement-time graph with slope θ_1 and for θ_2 possesses velocities v_1 and v_2

$$\text{respectively, then } \frac{v_1}{v_2} = \frac{\tan \theta_1}{\tan \theta_2}$$

10. For a particle accelerated for a time t_1 with acceleration a_1 and for t_2 with acceleration a_2 , then average

$$\text{acceleration is } a_{av} = \frac{a_1t_1 + a_2t_2}{t_1 + t_2}$$

11. Scalar or Dot Product of Unit Vectors

$$(i) \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \quad (ii) \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

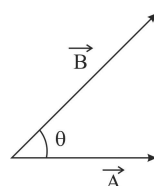
where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are orthogonal unit vectors.

12. Vector Product or Cross Product of Unit Vectors

$$(i) \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

$$(ii) \hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$$

13. $\vec{A} \cdot \vec{B} = AB \cos \theta$

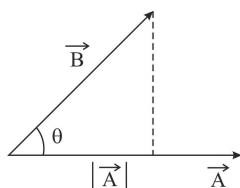


$$\text{and } \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

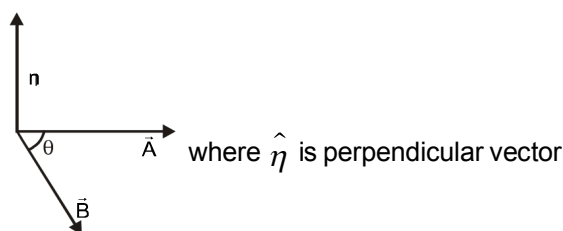
θ = angle between A and B

14. Projection of vector $\vec{B}$ in the direction of vector $\vec{A}$

$$B \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}|}$$



15. (i) $\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$

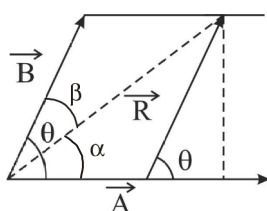


(ii) If $|\vec{V}_1 + \vec{V}_2| = |\vec{V}_1 - \vec{V}_2|$ and v_2 is finite, then v_1 and v_2 are perpendicular to each other.

16. Parallelogram Law of Vector Addition

$$(i) R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$(ii) \tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$



where A and B are magnitudes of vectors,

R = magnitude of resultant vectors,

θ = angle between vectors A and B,

α = angle between resultant vector R and vector A.

17. Projectile Motion

Suppose a particle is projected from origin on horizontal ground with a velocity u at an angle θ from the horizontal

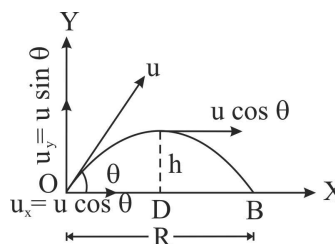
- (i) Trajectory of the projectile

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

(ii) Time of flight, $T = \frac{2u \sin \theta}{g}$

(iii) Range, $R = \frac{u^2 \sin 2\theta}{g}$

and $R_{\max} = \frac{u^2}{g}$ (Here $\theta = 45^\circ$)



(iv) Height attained, $H = \frac{u^2 \sin^2 \theta}{2g}$

(v) Kinetic energy of the projectile at the highest point of trajectory $K' = K \cos^2 \theta$

where K = Kinetic energy at the point of launch

(vi) A projectile is thrown with a speed v at an angle θ has a range R on the surface of earth for same value of v and θ its range on the surface of the moon will be $6R$.

18. Circular Motion

(i) Angular velocity, $\vec{\omega} = \frac{d\vec{\theta}}{dt}$

(ii) Angular acceleration, $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

(v) Centripetal force, $F = \frac{mv^2}{r} = mr\omega^2$

19. * Force, $F = \frac{dp}{dt} = ma$

Where a = acceleration produced in a body.

* **Weight in a Lift**

(i) Apparent weight, $R = m(g + a)$

where a = upward acceleration of the lift.

(ii) $R = m(g - a)$

where a = downward acceleration of the lift.

(iii) If free fall under the action of gravity, $a = g$

$$\therefore R = 0$$

(iv) When lift is at rest or moving uniformly, $a = 0$

$$\therefore R = mg$$

20. Recoil Velocity of Gun, $v_2 = -\frac{m_1 v_1}{m_2}$

where m_2 = mass of gun, m_1 = mass of bullet and v_1 = velocity of bullet.

Impulse

(i) Impulse = $F \times t$

where F = force and t = time.

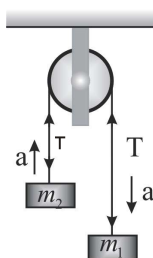
(ii) Impulse = $m(v - u)$

where v = final velocity

and u = initial velocity

21. Pulleys

Unequal mass ($m_1 > m_2$) suspended from a pulley

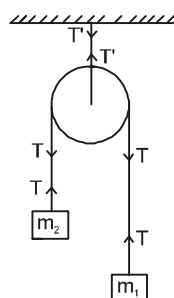


(a) Acceleration, $a = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) g$

(b) Tension, $T = \left(\frac{2m_1 m_2}{m_1 + m_2} \right) g$

* If $M_1 > M_2$ then accn of the block $a = \frac{(m_1 - m_2)g}{(m_1 + m_2)}$

Tension in the string $T = \frac{2m_1 m_2 g}{(m_1 + m_2)}$

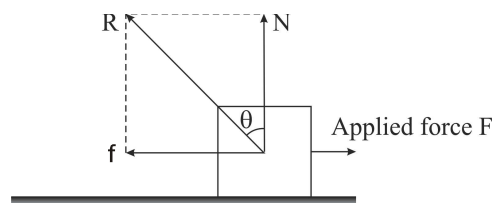


*Tension in the string by which pulley attached

$T' = \frac{4m_1 m_2 g}{(m_1 + m_2)}$

22. Friction

(i) Angle of friction, $\tan \theta = \mu_s$



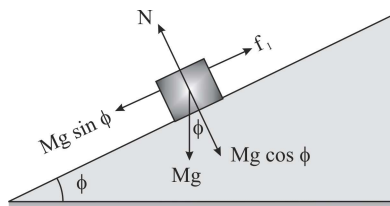
(ii) Coefficient of friction $\mu = \frac{f}{N}$

Where f = force of friction,

N = normal reaction.

(iii) Angle of repose

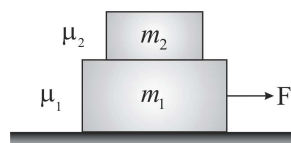
Angle of friction (θ) = angle of repose ϕ



(iv) Acceleration of a body down a rough plane

$a = g(\sin \phi - \mu \cos \phi)$

(v) When force is applied on lower block common acceleration, $a = \mu g$



23. * Work

$W = \vec{F} \cdot \vec{s} = Fs \cos \theta$

where F = constant force,

s = displacement of body,

θ = angle between displacement and force.

* Energy

(i) Kinetic energy, $KE = \frac{1}{2}mv^2$

where m = mass of the body,

v = velocity of the body.

(ii) Potential energy

(a) Elastic potential energy, $U = \frac{1}{2} kx^2$

Where k = spring constant

x = compression or expansion in spring

(b) Gravitational potential energy,

$$U = -Gm_1m_2/r$$

(c) Electrostatic potential energy,

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r}$$

* Power

(i) $P_{av} = \frac{W}{\Delta t}$

(ii) $P_{inst} = \frac{dW}{dt}$

(iii) $P = \vec{F} \cdot \vec{v}$

* Collision

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

(when linear momentum is conserved)

where m_1 and m_2 = masses of the body which undergo collision,

u_1 = initial velocity of body of mass m_1 ,

u_2 = initial velocity of body of mass m_2 ,

v_1 = final velocity of body of mass m_1 ,

v_2 = final velocity of body of mass m_2 .

(i) Inelastic collision

$$(a) v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2} \right) u_1 + \left(\frac{(1+e)m_2}{m_1 + m_2} \right) u_2$$

$$(b) v_2 = \left(\frac{(1+e)m_2}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - em_1}{m_1 + m_2} \right) u_2$$

(ii) Elastic collision

$$(a) \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$(b) v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2} + \frac{2m_2u_2}{m_1 + m_2}$$

$$(c) v_2 = \frac{2m_1u_1}{m_1 + m_2} + \frac{(m_2 - m_1)u_2}{m_1 + m_2}$$

(iii) Coefficient of Restitution,

$$e = \frac{|\vec{v}_2 - \vec{v}_1|}{|\vec{u}_1 - \vec{u}_2|}$$

(iv) Energy Loss in Collision

(i) Elastic collision, for first (incident) particle

$$\frac{K_f}{K_i} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2$$

$$\frac{\Delta K_{lost}}{K_i} = \frac{4m_1m_2}{(m_1 + m_2)^2}$$

$$\frac{\Delta K_{lost}}{K_i} = 100\% \quad (\text{if } m_1 = m_2)$$

(ii) Inelastic collision, loss in KE

$$K_{lost} = \frac{1}{2} \frac{m_1m_2}{m_1 + m_2} (u_1 - u_2)^2 (1 - e^2)$$

24. Equation of Rotational Motion

$$(\omega = \omega_0 + \alpha t), (\theta = \omega_0 t + \frac{1}{2} \alpha t^2), (\omega^2 - \omega_0^2 = 2\alpha \theta)$$

* Centre of Mass

(i) Centre of mass of N particle system

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + \dots + m_nx_n}{m_1 + m_2 + \dots + m_n}$$

where $x_1, x_2, \dots, x_n$ are the distances from the centre of masses $m_1, m_2, \dots, m_n$.

(ii) Velocity of centre of mass

$$\vec{v} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2 + \dots + m_n\vec{v}_n}{m_1 + m_2 + \dots + m_n}$$

where $v_1, v_2, \dots, v_n$ be velocities of the particles having masses $m_1, m_2, \dots, m_n$.

(iii) Acceleration of the centre of mass

$$\vec{a} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots + m_n \vec{a}_n}{m_1 + m_2 + \dots + m_n}$$

where $a_1, a_2, \dots, a_n$ be accelerations of the particles

having masses $m_1, m_2, \dots, m_n$.

(iv) Momentum of centre of mass

$$\vec{p} = \frac{m_1 \vec{p}_1 + m_2 \vec{p}_2 + \dots + m_n \vec{p}_n}{m_1 + m_2 + \dots + m_n}$$

25. Moment of Inertia

(i) MI of n particle system

$$I = \sum_{i=1}^n m_i r_i^2$$

(ii) MI of a body $I = MK^2$

Where M = total mass, K = radius of gyration.

Kinetic Energy of Rotation $K_R = \frac{1}{2} I \omega^2$

Kinetic Energy of Rolling Body

(i) KE of rolling body,

$$E_{\text{rolling}} = E_{\text{translation}} + E_{\text{rotational}}$$

$$E_{\text{rolling}} = \frac{1}{2} m v^2 + \frac{1}{2} m K^2 \omega^2$$

(ii) Total kinetic energy of a rolling body

$$= \frac{1}{2} m v^2 \left(1 + \frac{K^2}{r^2} \right)$$

Rolling of a Body on an Inclined Plane without Slipping

(i) Acceleration of the body $a = \frac{g \sin \theta}{1 + \frac{K^2}{r^2}}$

(ii) Velocity of the body, when it reaches the bottom

$$v = \sqrt{\frac{2gh}{1 + \frac{K^2}{r^2}}} = \sqrt{\frac{2gl \sin \theta}{1 + \frac{K^2}{r^2}}}$$

(iii) Time taken by a rolling body to reach the bottom

$$t = \frac{1}{\sin \theta} \sqrt{\frac{2h \left(1 + \frac{K^2}{r^2} \right)}{g}}$$

26. Torque

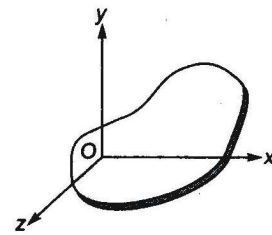
(i) torque of a body, $\vec{\tau} = \vec{r} \times \vec{F}$

(ii) Work done by a body, $W = \vec{\tau} \cdot \vec{d\theta}$

(iii) Power, $P = \vec{\tau} \cdot \vec{\omega} = \tau \cdot 2\pi n$

(iv) Power of torque $= \tau \times \frac{2\pi}{T}$

Theorem of Perpendicular Axis,



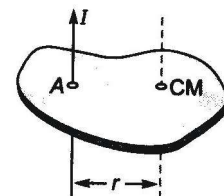
$$I_Z = I_X + I_Y$$

where I_Z = MI of the body about axis perpendicular to the plane of lamina,

I_X and I_Y = MI of the lamina about two mutually perpendicular axis of lamina.

Theorem of Parallel Axis,

$$I = I_{CM} + Mr^2$$



where I = MI of the body about an axis passing through any point,

I_{CM} = MI of the body about an axis passing through centre of mass of the body,

M = Mass of the body,

r = Distance between two parallel axis.

27. Angular Momentum

$$\vec{L} = I \vec{\omega} \quad \& \quad \vec{L} = \vec{r} \times \vec{p}$$

Law of conservation of angular momentum

$$I_1 \omega_1 = I_2 \omega_2$$

* The angular momentum of electron in nth orbit is given

$$\text{by } L = \frac{nh}{2\pi}$$

28. Newton's Law of Gravitation

$$F = \frac{Gm_1m_2}{r^2}$$

* Acceleration due to Gravity

$$g = \frac{GM}{R^2} \quad \text{weight} \quad W = mg$$

* Variation of g

(i) due to shape $g_{\text{equator}} < g_{\text{pole}}$

(ii) due to rotation of earth

$$g_{\text{pole}} = \frac{GM}{R^2} \quad (\text{no effect})$$

$$g_{\text{equator}} = \frac{GM}{R^2} - \omega^2 R$$

$$g_{\text{equator}} < g_{\text{pole}}$$

$$\omega^2 R = 0.034 \text{ ms}^{-2}$$

$$\text{If } \omega = 17\omega_0 \text{ or duration of day } T = \frac{24}{17} = 1.4 \text{ h}$$

then objects would float on equator.

(iv) Variation at depth d

$$g' = g \left(1 - \frac{d}{R} \right) \quad \frac{g - g'}{g} = \frac{d}{R}$$

* Gravitational Field

$$(i) \vec{g} = \vec{F} / m_0 \quad (ii) \vec{g} = -\frac{GM}{r^2} \hat{r}$$

(Outside)

$$(iii) \vec{g} = -\frac{GM}{R^3} r \cdot \hat{r} \quad (\text{Inside})$$

* Gravitational Potential Energy of Mass m

$$U(r) = \frac{-GMm}{r}$$

$$\text{At the surface of the earth } U_0 = -\frac{GMm}{R}$$

* Escape Velocity

$$(i) v_e = \sqrt{\frac{2GM}{R}} \quad (ii) v_e = \sqrt{2gR}$$

$$(iii) v_e = \sqrt{gD} \quad (iv) v_e = R \sqrt{\frac{g\pi G\rho}{3}}$$

29. Orbital Velocity of Satellite

$$(i) v_0 = \sqrt{\frac{GM}{r}} \quad (ii) v_0 = v_e \sqrt{\frac{M}{2(R+h)}}$$

$$(iii) v_0 = \frac{v_e}{\sqrt{2}} \quad (\text{for } h \ll R)$$

* Time Period of Satellite

$$(i) T = 2\pi \sqrt{\frac{(R+h)^3}{GM}} \quad (ii) T = 2\pi \sqrt{\frac{R}{g}} \quad (h \ll R)$$

* Energy of Satellite

$$(i) \text{ Kinetic energy } K = \frac{GMm}{2r}$$

$$(ii) \text{ Potential energy } U = -\frac{GMm}{r}$$

$$(iii) \text{ Total energy } E = -\frac{1}{2} \frac{GMm}{r} = \frac{U}{2} = -K$$

$$(iv) \text{ Binding energy } BE = -E = \frac{1}{2} \frac{GMm}{r}$$

* Geosynchronous Satellite

$$(i) T = 24 \text{ h} \quad (ii) T^2 = \frac{4\pi^2}{GM} (R+h)^3$$

$$(iii) h = \left(\frac{GMT^2}{4\pi^2} \right)^{1/3} - R \quad (iv) h = 36000 \text{ km}$$

* Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of escape

$$\text{velocity } \frac{v_A}{v_B} \text{ is } 2.$$

30. Kepler's Laws

(i) Law of orbits : orbits are elliptical.

(ii) Law of areas : equal area swept in equal time

$$\frac{\Delta A}{\Delta t} = \frac{L}{2m} \quad v_{\text{max}} r_{\text{min}} = v_{\text{min}} r_{\text{max}}$$

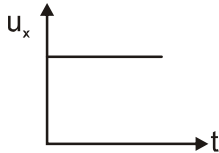
(iii) Law of periods

$$T^2 \propto r^3 \quad T^2 = \left(\frac{4\pi^2}{GM} \right) r^3$$

- * The acceleration due to gravity at a depth d where g is acceleration due to gravity on the surface of earth, will

$$\text{be } g \left(1 - \frac{d}{R} \right).$$

- * The graph between horizontal velocity and time t in a projectile motion is given by :



$$\text{Stress} = \frac{\text{Deforming force}}{\text{Cross-sectional area}}$$

31. Strain

(i) Tensile or longitudinal strain = $\frac{\Delta L}{L}$

(ii) Shearing strain = ϕ

(iii) Volume strain = $\frac{\Delta V}{V}$

- * Hooke's Law

(i) For stretching $\text{stress} = Y \times \text{strain}$

$$Y = \frac{FL}{A(\Delta L)}$$

(ii) For shear $\text{stress} = \eta \times \text{strain}$

$$\eta = \frac{F}{A\phi}$$

(iii) for volume elasticity $\text{stress} = K \times \text{strain}$

$$K = -\frac{P}{(\Delta V/V)}$$

* Compressibility $\beta = \frac{1}{K}$

- * Elongation of a Wire due to its Own Weight

$$\Delta L = \frac{1}{2} \frac{MgL}{YA} = \frac{1}{2} \frac{L^2 \rho g}{Y}$$

32. Bulk Modulus of Ideal Gas

(i) $K_{\text{isothermal}} = p$ (ii) $K_{\text{adiabatic}} = \gamma p$

33. Thermal stress

(i) Thermal stress = $Y\alpha(\Delta t)$

(ii) Force = $YA\alpha(\Delta t)$

34. Work Done in Stretching

(i) $W = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$

(ii) $W = \frac{1}{2} \times Y(\text{strain})^2 \times \text{volume}$

(iii) $W = \frac{1}{2} \times \frac{(\text{stress})^2}{Y} \times \text{volume}$

(iv) Potential energy stored

$$U = \frac{1}{2} \times (\text{stress}) \times (\text{strain}) \times \text{volume}$$

(v) Potential energy stored per unit

volume $U = \frac{1}{2} \times (\text{stress}) \times (\text{strain})$

35. Loaded Beam

(i) Depression $\delta = \frac{WI^3}{4Ybd^3}$ (rectangular)

$$\delta = \frac{WI^3}{12Y\pi d^3}$$
 (cylindrical)

- * Poisson's Ratio

(i) Lateral strain = $\frac{-\Delta D}{D} = \frac{-\Delta r}{r}$

(ii) Longitudinal strain = $\frac{\Delta L}{L}$

$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$-1 < \sigma < 0.5$ (theoretical limit)

$\sigma = 0.2 - 0.4$ (experiments)

36. Principle of Continuity

$$Av = \text{Constant}$$

where A = area of cross-section,
 v = velocity of flow.

* Bernoulli's Theorem

$$p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

where p = pressure

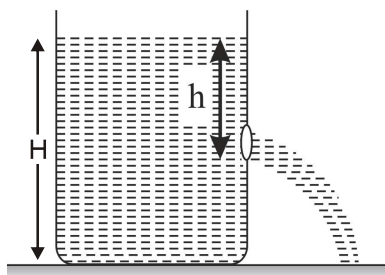
$$\frac{1}{2}\rho v^2 = \text{kinetic energy per unit volume}$$

$$\rho gh = \text{potential energy per unit volume}$$

* Velocity of Efflux

$$v = \sqrt{2gh}$$

where h = height of the hole.



The time taken by the liquid to reach the base level in above case

$$t = \sqrt{\frac{2(H-h)}{g}}$$

where H = height of vessel.

37. Viscous Force

$$F = -\eta A \frac{dv}{dy}$$

where η = coefficient of viscosity,
 A = area of layer of fluid in contact,

$\frac{dv}{dy}$ = velocity gradient between the layers.

38. Poiseuille's Formula
$$\frac{dQ}{dt} = \frac{\pi p R^4}{8\eta L}$$

where L = length of the capillary tube,
 R = radius of the capillary tube,
 p = pressure difference across the tube.

* Terminal Velocity,
$$v_T = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta}$$

where r = radius of the body,
 ρ = density of material of the body,
 σ = density of fluid,
 η = coefficient of viscosity.

40. Surface Tension,
$$T = \frac{F}{L}$$

where F = force per unit length.

* Force due to Surface Tension

(i) Force required to lift a thin wire from surface of the water
$$F = T \times 2l$$

where l = length of thin wire.

(ii) Force required to lift a rectangular plate from surface of the water
$$F = 2T(l + b)$$

where l = length of rectangular plate,
 b = breadth of rectangular plate.

(iii) Force required to lift a circular plate from surface of water
$$F = T \times 2\pi r$$

(iv) Force required to lift a ring from surface of water
$$F = T(2r_1 + 2r_2)$$

where r_1 = inner radius, r_2 = outer radius.

(v) Force required to separate two glass plates of area A containing a liquid film of thickness d between them,

$$F = \frac{2TA}{d}$$

41. Work done to Increase the Area of Surface Film by A

$$W = T \times A$$

* Work done in blowing a soap bubble of radius r
$$W = 8\pi r^2 T$$

* Work done in splitting a liquid drop of radius r into n identical drops
$$W = 4\pi T r^2 (n^{1/3} - 1)$$

* Work done in combining n identical drop of radius r to form a big drop of radius R

$$W = 4\pi T (nr^2 - R^2) = 4\pi T r^2 (n - n^{2/3})$$

42. Capillary Rise or Fall of a Liquid

$$h = \frac{2T \cos \theta}{r \rho g}$$

where h = height of liquid column in capillary,
 T = surface tension of the liquid,
 θ = angle of contact,
 r = radius of the capillary tube and d is the density of the liquid.

For pure liquid $\theta = 0^\circ$ $h = \frac{2T}{r \rho g}$

(a) If $\theta < 90^\circ$, then there will be a capillary rise.

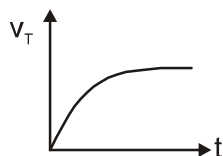
(b) If $\theta = 90^\circ$, there will be neither rise nor fall.

(c) If $\theta > 90^\circ$, then there will be a depression.

43. When temperature of water increases its viscosity decreases.

* Variation of terminal velocity v_T and time for a long

column of a viscous fluid will be :



44. Excess Pressure

(i) Excess pressure in a liquid drop of radius, r

$$p = \frac{2T}{r}$$

(ii) excess pressure in a soap bubble of radius r

$$p = \frac{4T}{r}$$

(iii) Excess pressure in a soap bubble of radius r ,

when it is inside a liquid $p = \frac{2T}{r}$

45. If two soap bubbles of radii r_1 and r_2 ($r_1 > r_2$) are in contact with each other, then the radius of curvature of

their interface $r = \frac{r_1 r_2}{r_1 - r_2}$

* If two soap bubble of radii r_1 and r_2 combine in vacuum to form a single bubble of radius r under isothermal condition, then

$$r = \sqrt{r_1^2 + r_2^2}$$

*. If a number of small droplets of water, all of the same radius r , coalesce to form a single drop of radius R , then

$$\text{Rise in temperature} = \frac{3T}{J \rho s} \left(\frac{1}{r} - \frac{1}{R} \right)$$

where s = specific heat, ρ = density.

* The radii of two limbs of a U-tube are r_1 and r_2 . A liquid of density ρ and surface tension T is poured in it. The difference between the levels of the liquid.

$$h = \frac{2T}{\rho g} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

46. (i) $\frac{C}{100} = \frac{F - 32}{180}$

$$\Rightarrow F = \frac{9}{5}C + 32 \Rightarrow C = \frac{5}{9}(F - 32)$$

(ii) $K = C + 273$ (iii) $K = \frac{5}{9}(F - 459.4)$

47. (i) Triple point of water $T_3 = 273.16 K$

(ii) Absolute zero $0K \equiv -273.15^\circ C$

48. (i) Isobaric, $p = \text{constant}$; $\frac{V}{T} = \text{constant}$

(ii) Isochoric (isometric)

$$V = \text{constant}; \frac{p}{T} = \text{constant}$$

(iii) Isothermal $T = \text{constant}$; $pV = \text{constant}$

(iv) Adiabatic $pV^\gamma = \text{constant}$; $TV^{\gamma-1} = \text{constant}$;

$$p^{1-\gamma} T^\gamma = \text{constant}$$

*. (i) Isobaric slope : Zero

(ii) Isochoric slope : Infinite

(iii) Isothermal : Slope = $-\frac{p}{V}$

(iii) Adiabatic : Slope = $-\frac{\gamma p}{V}$

48. (i) Isobaric $W = p(V_1 - V_2)$

(ii) Isochoric $W = 0$

(iii) Isothermal $W = nRT \log_e (V_2 / V_1)$

$$W = nRT (2.303) \log_{10} (V_2 / V_1)$$

$$W = nRT (2.303) \log_{10} (p_1 / p_2)$$

(iv) Adiabatic: $W = \frac{nR(T_1 - T_2)}{\gamma - 1}$

(iv) In expansion

$$W_{\text{adiabatic}} < W_{\text{isothermal}} < W_{\text{isobaric}}$$

(vi) In compression

$$W_{\text{adiabatic}} > W_{\text{isothermal}} > W_{\text{isobaric}}$$

Work done depends on path on pV diagram.

49. (i) Isobaric $Q = nC_p \Delta T$

(ii) Isochoric $Q = nC_v \Delta T$

(iii) Isothermal $Q = W = nRT \log_e (V_2 / V_1)$ etc.

(iv) Adiabatic $Q = 0$

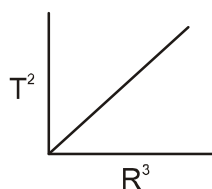
50. (i) Isobaric $\Delta U = nC_v \Delta T$

(ii) Isochoric $\Delta U = nC_v \Delta T$

(iii) Isothermal $\Delta U = 0$

(iv) Adiabatic $\Delta U = -W = \frac{nR(T_2 - T_1)}{\gamma - 1}$

51. The graph between motion of planet and time period of revolution is given by :



52. Work done per mol in an isothermal change is

$$W = RT \log_e \frac{V_2}{V_1}$$

* (i) Isothermal bulk modulus $K = p$

(ii) Adiabatic bulk modulus $K = \gamma p$

* (i) Work $\Delta W =$ area enclosed in pV diagram

$$\Delta U = 0 \quad \Delta Q = \Delta W$$

* (i) $U = \frac{3}{2} RT$ (one mole)

(ii) $U = \frac{3}{2} nRT$ (n mole)

(iii) Change $\Delta U = \frac{3}{2} nR \Delta T$

54. $\Delta Q = \Delta U + \Delta W$, $\Delta U = \Delta Q - \Delta W$

Both ΔQ , ΔW depends on path, but ΔU does not depend on the path.

(i) Isothermal $\Delta Q = \Delta W$, $\Delta U = 0$

(ii) Adiabatic $\Delta Q = 0$, $\Delta U = -\Delta W$

(iii) Isochoric $\Delta W = 0$, $\Delta Q = \Delta U = nC_v \Delta T$

(iv) Isobaric $\Delta Q = \Delta U + \Delta W$,

$$\Delta Q = nC_p \Delta T,$$

$$\Delta U = nC_v \Delta T,$$

$$\Delta W = nR \Delta T$$

(v) Cyclic $\Delta U = 0$, $\Delta Q = \Delta W$

(vi) Free expansion $\Delta U = 0$, $\Delta Q = 0$, $\Delta W = 0$

(vii) Specific heat $s = \frac{\Delta Q}{m \Delta T}$

(viii) Change of state $\Delta Q = mL$

(ix) Work converts into heat

$$W = JH \quad (J = 4.1860 \text{ J} \cdot \text{cal}^{-1})$$

(x) Memorable point

$$2^{0.4} = 1.32, \quad 2^{1.4} = 2 \times 2^{0.4} = 2.64$$

55. (i) Efficiency of engine, $\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$

where Q_1 = amount of heat absorbed by the working substance from the source at T_1 K.

 Q_2 = amount of heat rejected to the sink at T_2 K

(ii) Coefficient of performance,

$$\beta = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

where W = net amount of external work done.

* Entropy $S = \frac{dQ}{T}$

where T = absolute temperature,

 m = mass of substance,

56. Pressure Exerted by an Ideal Gas

$$\text{Pressure } p = \frac{1}{3} \frac{m n c^2}{V}$$

where m = mass of each molecule,
 n = number of molecules in the enclosure,
 c = root mean square velocity of molecules,
 V = volume of container enclosed mass M of gas.

* According to the kinetic theory of gases total energy of a gas is equal to kinetic energy.

57. (i) Boyle's law $p \propto \frac{1}{V}$ ($\because n$ & T = constant)

(ii) Charles' law $V \propto T$ ($\because n$ & p = constant)

(iii) Avogadro law $\frac{V_1}{N_1} = \frac{V_2}{N_2}$

(iv) Ideal gas equation $pV = nRT$

where p = pressure of gas,
 V = volume of gas,
 n = no. of moles
 R = ideal gas constant,
 T = temperature of gas,

(v) Dalton's law of partial pressures

$$p_T = p_1 + p_2 + p_3 + \dots$$

where p_T = total pressure,
 $p_1, p_2, \dots$ = partial pressure of the number of gases.

58. (i) For monoatomic gas = $\frac{3}{2} kT$ (per molecule)
 and $\frac{3}{2} RT$ (per mole)

(ii) For diatomic gas = $\frac{5}{2} kT$ (per molecule)
 and $\frac{5}{2} RT$ (per mole)

(iii) For triatomic gas = $3kT$ (per molecule)
 and $3RT$ (per mole)

59. (iv) Emissive power $E = \frac{Q}{A \cdot t}$

* $\left(\frac{e_\lambda}{a_\lambda} \right) = \left(\frac{e_\lambda}{a_\lambda} \right) = \dots = E_\lambda$

* $E = \sigma T^4$ ($\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)

(i) $E = \sigma (T^4 - T_0^4)$ (for a black body)

(ii) $e = \epsilon \sigma (T^4 - T_0^4)$ (for a body)

* (i) $\frac{d\theta}{dt} = \frac{\epsilon A \sigma}{ms} (\theta^4 - \theta_0^4)$

(ii) $\frac{d\theta}{dt} = \frac{\epsilon A \sigma}{V \rho s} (\theta^4 - \theta_0^4)$

* Newton's Law of Cooling

(ii) $\frac{\theta_1 - \theta_2}{t} = K \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$

* $\lambda_m T = b$ ($b = 2.9 \times 10^{-3} \text{ m-K}$)

60. (i) $v = n\lambda$ (ii) $n = 1/\lambda$

(ii) $v = \sqrt{\frac{T}{m}}$, m = mass per unit length

(iii) $v = \sqrt{\frac{T}{\pi r^2 d}}$

(iv) In rods $v = \sqrt{\frac{Y}{\rho}}$

In liquids $v = \sqrt{\frac{B}{\rho}}$

In gases $v = \sqrt{\frac{\gamma P}{\rho}}$

(vii) $\{v_{\text{sound}}/v_{\text{rms}} = \sqrt{\gamma/3}\}$

61. (i) $y = a \sin(\omega t - kx)$,

(ii) Wave velocity $v = \omega/k$

(iii) Particle velocity $v_{\text{particle}} = \frac{\delta y}{\delta t}$

(iv) Maximum particle velocity $(v_{\text{particle}})_{\text{max}} = \omega a$

(v) Strain in medium = $-\frac{\delta y}{\delta x}$

(vii) When a longitudinal wave propagate through a material medium then energy and linear momentum always remains constant.

(viii) $I = 2\pi^2 \rho v n^2 a^2$ $I \propto n^2 a^2$

(ix) Intensity level in decibel = $10 \log \left(\frac{I}{I_0} \right)$

62 (i) Resultant amplitude

$$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$$

(ii) Resultant intensity

$$I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi$$

(iii) Constructive interference

$$\phi = 2n\pi \quad a_{\max} = a_1 + a_2$$

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

(iv) Destructive interference

$$\phi = (2n - 1)\pi \quad a_{\min} = a_1 - a_2$$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

(v) Beat frequency = $n_1 - n_2$

63. (i) $y = -2a \sin kx \cos \omega t$ (ii) $y = 2a \cos kx \sin \omega t$

(iii) Vibrations of a string

$$n = \frac{1}{2\ell} \sqrt{\frac{T}{m}} \quad (\text{fundamental})$$

$$n_p = \frac{p}{\lambda} \sqrt{\frac{T}{m}} \quad (p^{\text{th}} \text{ harmonic})$$

The ratio of successive harmonics frequencies

$$n_1 : n_2 : n_3 : \dots = 1 : 2 : 3 : \dots$$

$$m = \pi r^2 d$$

(iv) Closed pipe Fundamental $n_1 = v / 4L$

Only odd harmonics present

$$n_1 : n_2 : n_3 : \dots = 1 : 3 : 5 : \dots$$

(v) Open pipe Fundamental $n_1 = v / 2L$

Both even and odd harmonics are present

$$n_1 : n_2 : n_3 = 1 : 2 : 3 : \dots$$

(vi) Both ends open, clamped in middle

(a) Fundamental $n_1 = v / 2l$

(b) First overtone $n_2 = 3n_1$

(c) Ratio of frequencies $n_1 : n_2 : n_3 = 1 : 3 : 5 : \dots$

(vii) One end clamped

(a) Fundamental $n_1 = v / 4l$

(b) First overtone $n_2 = 3n_1$ etc.

(c) Ratio of frequencies

$$n_1 : n_2 : n_3 = 1 : 3 : 5 : \dots$$

(ix) Tuning fork
$$n = \frac{t}{l^2} \sqrt{\frac{E}{\rho}}$$

where t = thickness, l = length of prong,

E = elastic constant and ρ = density

64. Doppler's Effect

(i) Observer stationary and source moving

(a) Source approaching

$$\lambda' = \frac{v - v_s}{v} \lambda, \quad n' = \frac{v}{v - v_s} n$$

(b) Source receding

$$n' = \frac{v}{v + v_s} n \quad \lambda' = \frac{v + v_s}{v} \lambda$$

(ii) Source stationary and observer moving

(a) Observer approaching source

$$n' = \frac{v + v_o}{v} n \quad \lambda' = \lambda$$

(b) Observer receding away from source

$$n' = \frac{v - v_o}{v} n \quad \lambda' = \lambda$$

(iii) Source and Observer both are moving

(a) S and O moving toward each other

$$n' = \frac{v + v_o}{v - v_s} n$$

(b) S and O moving away from each other

$$n' = \frac{v - v_o}{v + v_s} n$$

(c) S and O in same direction, O ahead of S

$$n' = \frac{v - v_o}{v - v_s} n$$

(d) S and O in same direction, S ahead of O

$$n' = \frac{v + v_o}{v + v_s} n$$

65. (i) $F = \frac{kq_1q_2}{r^2}$ where $k = \frac{1}{4\pi\epsilon_0}$,

q_1 and q_2 = charge of both particles and
 r = distance between both charges.

(ii) $F_{\text{medium}} = \frac{F_{\text{vacuum}}}{K}$

(i) $F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$

$$(ii) \tan \alpha = \frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta}$$

$$66. (i) \vec{E} = \frac{\vec{F}}{q_0} \quad (ii) \vec{E} = k \frac{q}{r^2} \hat{r}$$

$$67. (i) E = \sqrt{E_1^2 + E_2^2 + 2E_1E_2 \cos \theta}$$

$$(ii) \tan \alpha = \frac{E_2 \sin \theta}{\{E_1 + E_2 \cos \theta\}}$$

(iii) Position of zero field
(Like charges separated by distance R)

$$x = \frac{R}{\{1 + \sqrt{q_2 / q_1}\}}$$

$$68. (i) E = \frac{qx}{4\pi\epsilon_0 (a^2 + x^2)^{3/2}}$$

$$(ii) E_{\text{centre}} = 0$$

$$(iii) \omega = \sqrt{eq / 4\pi\epsilon_0 ma^3}$$

69. (i) Dipole moment $p = qd$

(ii) Field at an axial point

$$E = \frac{k \cdot 2pr}{[r^2 - (d/2)^2]^2} \quad E \approx \frac{k2p}{r^3} \quad (\text{if } r \gg d/2)$$

(iii) Field at an equatorial point

$$E = \frac{kp}{\{r^2 + (d/2)^2\}^{3/2}} \quad E = \frac{kp}{r^3}$$

$$E_{\text{axial}} / E_{\text{equa.}} = 2/1$$

(v) Torque on a dipole $\vec{\tau} = \vec{p} \times \vec{E}$

$$\tau = pE \sin \theta$$

Potential energy of a dipole

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$$

Work done in rotating a dipole moment from angle θ_1 to angle θ_2

$$W = U_2 - U_1 = pE (\cos \theta_1 - \cos \theta_2)$$

$$70. (i) \text{ Line charge } E = \frac{k2\lambda}{r} \quad E = \frac{\lambda}{2\pi\epsilon_0 r}$$

(ii) Cylindrical charge

$$(a) \text{ Outside } E = \frac{\lambda}{2\pi\epsilon_0 r}, E \propto \frac{1}{r}$$

$$(b) \text{ At the surface } E = \frac{\lambda}{2\pi\epsilon_0 R},$$

$$(c) \text{ Inside } E = \frac{\lambda r}{2\pi\epsilon_0 R^2}, E \propto r$$

$$(iii) \text{ Plane sheet } E = \frac{\sigma}{2\epsilon_0}$$

$$(iv) \text{ Conducting sheet } E = \frac{\sigma}{\epsilon_0}$$

(v) Field of charged spherical shell

$$(a) \text{ Outside } E = \frac{kq}{r^2}$$

$$(b) \text{ At the surface } E = \frac{kq}{R^2}$$

$$(c) \text{ Inside } E = 0$$

(vi) Field of uniformly charged non-conducting sphere

$$(a) \text{ Outside } E = \frac{kq}{r^2}, E \propto \frac{1}{r^2}$$

$$(b) \text{ At surface } E = \frac{kq}{R^2}$$

$$(c) \text{ Inside } E = \frac{kqr}{R^3}, E \propto r$$

$$71. \frac{dF}{dA} = \frac{\sigma^2}{2\epsilon_0}$$

Soap-bubble problem

$$p_{\text{in}} - p_{\text{out}} = \frac{4T}{r} - \frac{\sigma^2}{2\epsilon_0}$$

$$\text{or } r = \frac{8\epsilon_0 T}{\sigma^2} \text{ and } \sigma = \sqrt{8\epsilon_0 T / r}$$

$$q = 8\pi r \sqrt{2\epsilon_0 r T}$$

72. (a) $V = W/q$ (b) $\vec{E} = \frac{-\nabla V}{\Delta r}$

(i) Point charge $V = k \frac{q}{r}$

(ii) Many charges $V = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3}$

(iii) Potential of a charged spherical shell

(a) Outside $V = \frac{kq}{r}$

(b) At the surface $V = \frac{kq}{R}$

(c) Inside $V = V_{\text{surface}} = \frac{kq}{R}$

(iv) Potential of uniformly charged sphere (non-conducting)

(a) Outside $V = \frac{kq}{r}$

(b) At the surface $V = \frac{kq}{R}$

(c) Inside $V = \frac{kq(3R^2 - r^2)}{2R^3}$

$$V_{\text{center}} = \frac{3}{2} \frac{kq}{R} = 1.5V_{\text{surface}}$$

(v) Common potential (Two spheres joined by thin wire)

$$V = \frac{k(Q_1 + Q_2)}{r_1 + r_2}$$

$$q_1 = \frac{r_1(Q_1 + Q_2)}{r_1 + r_2} = \frac{r_1 Q}{r_1 + r_2}, \quad q_2 = \frac{r_2 Q}{r_1 + r_2}$$

$$\frac{q_1}{q_2} = \frac{r_1}{r_2}, \quad \frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1}$$

73. Potential Energy

(i) Two charges $U = \frac{kq_1 q_2}{r}, \quad U = q_2 V_1$

(ii) Three charges $U = \frac{kq_1 q_2}{r_{12}} + \frac{kq_1 q_3}{r_{13}} + \frac{kq_2 q_3}{r_{23}}$

74. (i) Energy Density $u = \frac{1}{2} \epsilon_0 E^2$

(ii) $C = q/V$ (iii) $U = \frac{1}{2} CV^2$

(iv) $U = \frac{1}{2} QV$ (v) $U = \frac{1}{2} \frac{Q^2}{C}$

75. (i) Sphere $C = 4\pi\epsilon_0 R$ $C = K(4\pi\epsilon_0 R)$

(ii) Spherical conductor

$$C = \frac{4\pi\epsilon_0 r_a r_b}{r_b - r_a} \quad C = \frac{4\pi\epsilon_r \epsilon_0 r_a r_b}{r_b - r_a}$$

(iii) Parallel plate $C = \frac{\epsilon_0 A}{d}$ $C = \frac{K\epsilon_0 A}{d}$

76. Series (i) q same, $q_1 = q_2 = q_3 = q$

(ii) V different $V = q/C_1$ $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

For two capacitors in series $C = \frac{C_1 C_2}{(C_1 + C_2)}$

Parallel (i) V same, $V_1 = V_2 = V_3 = V$

(ii) q different, $q_1 = C_1 V$, etc.

(iii) $C = C_1 + C_2 + C_3$

Dielectric (i) Field inside dielectric $E_d = E/K$

(ii) Polarization charges on surface of dielectric

$$Q_p = Q(1 - 1/K), \quad \sigma_p = Q_p / A = \sigma (1 - 1/K)$$

(iii) Polarization vector $p = \sigma_p$, $p = \epsilon_0 \chi E_d$

Capacitance Formulae with Dielectric

(i) $C = \frac{\epsilon_0 A}{d - t \left(1 - \frac{1}{k}\right)} = \frac{K\epsilon_0 A}{Kd - t(K - 1)}$

(ii) For metal slab of thickness t $C = \frac{\epsilon_0 A}{d - t}$

(iii) For n plates with alternate plates connected,

$$C = \frac{(n-1)\epsilon_0 A}{d} \quad C = \frac{\epsilon_0 A}{\left(\frac{t_1}{k_1} + \frac{t_2}{k_2} + \frac{t_3}{k_3}\right)}$$

77. (i) $E = \frac{W}{Q}$ $I = \frac{E}{r + R}$ $V = E - Ir$

Series combination :

(ii) $R = R_1 + R_2 + R_3$
 $V = V_1 + V_2 + V_3$

Parallel :

(iii) $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$V =$ same across all R

(iv) Heating Effects of Current

$W = I V t$ $P = IV$ $P = I^2 R, P = \frac{V^2}{R}$

(v) $H = I^2 R t$ Joule $H = \frac{I^2 R t}{J}$ calorie

(vii) Electric Bulbs $R = \frac{V^2}{P}$

Parallel Combination $P = P_1 + P_2 + P_3$

Series Combination $\frac{1}{P} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3}$

Effect of Stretching $\frac{R_2}{R_1} = \frac{\ell_2}{\ell_1} \times \frac{A_1}{A_2}$

$I_1 A_1 = I_2 A_2, \frac{R_2}{R} = \left(\frac{\ell_2}{\ell_1}\right)^2, \frac{R_2}{R_1} = \left(\frac{A_1}{A_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^4$

78. Mixed Combination

(m rows with each containing n cells in series)

$I = \frac{nE}{\frac{nr}{m} + R}$ $I = \frac{mnE}{nr + mR}$

Kirchhoff's First Law $\sum I = 0$

Kirchhoff's Second Law

$\sum E - \sum IR = 0$, or $\sum E = \sum IR$

Wheatstone Bridge

$\frac{P}{Q} = \frac{R}{S}$ (for balanced bridge)

79 Moving Coil Galvanometer

$\tau = NIAB, \tau = k\theta$ $i = \left(\frac{k}{NAB}\right)\theta$

(i) Current sensitivity = $\frac{\theta}{I} ; \frac{NAB}{k}$

(ii) Voltage sensitivity = $\frac{\theta}{V} = \left(\frac{\theta}{I}\right)R = \frac{NAB}{kR}$

80. Ammeter

(i) Shunt resistance $S = \frac{I_g G}{(I - I_g)}$

(ii) Length of shunt wire $l = \frac{S\pi r^2}{\rho}$

(iii) Volt meter $R = \frac{V}{I_g} - G$

(ii) For converted voltmeter, $R_v = R + G$
 Potentiometer

(i) Potential gradient

$k = \frac{E_p}{L}, k = \frac{E_s}{l}$

(ii) Comparison of emf

$\frac{E_1}{E_2} = \frac{\ell_1}{\ell_2}, E_2 = \frac{\ell_2}{\ell_1} \cdot E_1$

(iii) Internal resistance

$r = \frac{E - V}{I}, e = \frac{E - V}{V} R, r = \frac{I_1 - I_2}{I_2} R$

81. Biot-Savart Law

$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$

(i) Field due to a long straight wire $B = \frac{\mu_0 I}{2\pi r}$

(iii) Magnetic moment of circular coil

$m = NIA$

Field : $B = \frac{\mu_0}{4\pi} \cdot \frac{2m}{x^3}$

(iv) Field due to an arc of current

$$B = \frac{\mu_0 I \theta}{4\pi R^2} \quad B = \frac{\mu_0 I \theta}{4\pi R}$$

(v) At centre of a semicircular coil $B = \frac{\mu_0 I}{4R}$

(vi) Field due to finite length of wire

$$B = \frac{\mu_0 I}{4\pi a} (\sin \alpha_1 + \sin \alpha_2)$$

(vii) field at the centre of a square loop

$$B = \frac{2\sqrt{2}\mu_0 I}{\pi l}$$

Ampere's Law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$ $\oint \vec{H} \cdot d\vec{l} = I$

(i) Field due to current in a cylindrical rod

$$\text{Outside : } B = \frac{\mu_0 I}{2\pi r}$$

$$\text{Surface : } B = \frac{\mu_0 I}{2\pi R}$$

$$\text{Inside : } B = \frac{\mu_0 I r}{2\pi R^2}$$

(ii) Solenoid

$$\text{Field inside : } B = \mu_0 n I$$

$$\text{Field at one end : } B = \frac{1}{2} \mu_0 n I$$

(iii) Force between two parallel conductor

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

(iv) Path of a moving charge in a magnetic field

(a) When $\vec{v}$ is perpendicular to $\vec{B}$

$$\text{Path = Circular} \quad r = \frac{mv}{qB} \quad v = \frac{qB}{2\pi m},$$

$$T = \frac{2\pi m}{qB}, \omega = \frac{qB}{m} \quad (\text{cyclotrom frequency})$$

(b) When angle between $\vec{v}$ to $\vec{B}$ is θ

Path = Helical

$$r = \frac{mv_{\perp}}{qB} = \frac{mv \sin \theta}{qB} \quad v = \frac{qB}{2\pi m}, T = \frac{2\pi m}{qB}$$

(v) Velocity of paricle at radius r

$$V = \frac{qBr}{m}$$

(vi) The maximum kinetic energy

$$K = \frac{1}{2} \frac{q^2 B^2 R^2}{m}$$

82. (i) If a charged particle moves along a circle under the action of magnetic and electric fields then this region of space may have $E=0$, $B \neq 0$

(ii) Magnetic Field Produced by a Moving Charge

$$(i) \quad \vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \vec{r}}{r^2}$$

$$(ii) \quad B = \frac{\mu_0}{4\pi} \frac{qv \sin \theta}{r}$$

Magnetic Field

$$(i) \quad B = \frac{F_{\max}}{qV}$$

Atomic magnetic Moments

$$(i) \quad \mu_L = -\frac{eL}{2m} \quad (ii) \quad \mu_S = -\frac{eS}{m}$$

$$(iii) \quad \mu_J = -g \frac{eJ}{2m}$$

$$\text{Intensity of Magnitization} \quad I = \frac{M}{V}$$

$$\text{Magnetizing Field H} \quad H = \frac{B}{\mu_0} - 1$$

$$(i) \quad \text{For vacuum} \quad H = \frac{B}{\mu_0}$$

$$(ii) \quad \text{For medium} \quad H = \frac{B}{\mu}$$

$$\text{Magnetic Susceptibility} \quad \chi = \frac{1}{H}$$

Permeability

$$(i) \quad \mu = \frac{B}{H} \quad (ii) \quad \mu_r = \frac{\mu}{\mu_0} \quad (iii) \quad \mu_0 = \frac{B}{B_0}$$

Other Relations

$$(i) \quad \mu = \mu_0 (1 + \chi) \quad (ii) \quad \mu_r = 1 + \chi$$

$$(iii) \quad B = B_0 (1 + \chi) \quad (iv) \quad B = \mu_0 (H + I)$$

83. Field Due to a Bar Magnet

(i) Axial point

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2Mr}{(r^2 - l^2)^2} \quad B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3}$$

(ii) Equatorial point

$$B = \frac{\mu_0}{4\pi} \cdot \frac{M}{(r^2 + l^2)^{3/2}} \quad B = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3}$$

(iii) At arbitrary point $B = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3} \sqrt{1 + 3 \cos^2 \theta}$

(iv) In uniform field $\vec{F} = 0$

$$\vec{\tau} = \vec{M} \times \vec{B} \quad \tau = MB \sin \theta$$

(v) Potential Energy $U = -MB \cos \theta = -\vec{M} \cdot \vec{B}$

Tangent Galvanometer

$$(i) \quad B = B_H \tan \theta ; I = K \tan \theta \quad (ii) \quad K = \frac{2rB_H}{\mu_0 n}$$

Magnetic Flux

$$(i) \quad d\phi = \vec{B} \cdot \vec{dA} = B dA \cos \theta$$

$$(ii) \quad \phi = \int \vec{B} \cdot \vec{dA} \quad (iii) \quad \phi = BA \cos \theta$$

$$(iv) \quad \oint \vec{B} \cdot \vec{ds} = 0 \quad (v) \quad \nabla \cdot \vec{B} = 0$$

Faraday's Laws

$$(i) \quad e \propto \frac{d\phi}{dt} \quad (ii) \quad e = - \frac{d\phi}{dt}$$

$$(iii) \quad I = \frac{e}{R} = - \frac{d\phi}{R dt} \quad (iv) \quad dq = - \frac{d\phi}{R}$$

$$(v) \quad q = \frac{(\phi_1 - \phi_2)}{R}$$

$$\text{Self Inductance } L = \frac{\phi}{I} \quad e = -L \frac{dI}{dt}$$

$$\text{Solenoid } L = \frac{\mu_0 N^2 A}{l}$$

$$L = \mu_0 n^2 A l \text{ (with } n = N/l) \quad L = \frac{\mu_r \mu_0 N^2 A}{l}$$

$$\text{Mutual Inductance } M = \frac{\phi_2}{I_1} \quad e_2 = -M \frac{dI_1}{dt}$$

$$\text{Two solenoids } M = \frac{\mu_0 N_s N_p A}{l_p}$$

84. Series and Parallel Combination

$$(i) \quad \text{Series } L = L_1 + L_2 \quad L = L_1 + L_2 \pm 2M$$

$$(ii) \quad \text{Parallel } L = \frac{L_1 L_2}{L_1 + L_2} \quad \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$$

$$M = K \sqrt{L_1 L_2}$$

Response of R, L and C

(i) **R only :**

Phase difference $\phi = 0$

$$E_0 = I_0 R, \quad E_{rms} = I_{rms} R$$

(ii) **L only :**

Current lags voltage, voltage, leads current by a phase $\pi/2$

$$E_0 = I_0 X_L; \quad E_{rms} = I_{rms} X_L \quad X_L = \omega L$$

(iii) **C only :**

The current leads voltages, the voltage lags current by a phase $\pi/2$

$$E_0 = I_0 X_C; \quad E_{rms} = I_{rms} X_C \quad X_L = \omega L$$

(iv) **Series LR :**

The current lags, the voltages leads by a phase ϕ ,

$$\phi = \tan^{-1} \left(\frac{X_L}{R} \right) \quad \cos \phi = \frac{R}{Z}, \quad \sin \phi = \frac{X_L}{Z}$$

$$Z = \sqrt{R^2 + (\omega L)^2} \quad E_0 = I_0 Z, \quad E_{rms} = I_{rms} Z$$

(vi) **Series RC**

$$\phi = \tan^{-1} \left(\frac{X_C}{R} \right)$$

Current leads, voltage lags

$$\cos \phi = \frac{R}{Z}, \quad Z = \sqrt{R^2 + \left(\frac{1}{\omega C} \right)^2}$$

$$E_0 = I_0 Z, \quad E_{rms} = I_{rms} Z$$

(vi) Series LRC

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

ϕ is positive for $X_L > X_C$

ϕ is negative for $X_L < X_C$

current inductive if $X_L > X_C$

circuit inductive if $X_L > X_C$

current leads if $X_L < X_C$

circuit capacitive if $X_L < X_C$

(vii) Resonance

$$\text{Resonance frequency } f_r = \frac{1}{2\pi\sqrt{LC}}$$

At resonance, $X_L = X_C$, $V_L = V_C$

$\phi = 0$, $Z = R$ (minimum)

$\cos \phi = 1$, $\sin \phi = 0$ $I_0 = \text{maximum} = E_0 / R$

It is a band selector circuit.

$$85. \text{ Power in AC Circuits } P_{av} = \frac{E_0 I_0 \cos \phi}{2}$$

$$P_{av} = E_{rms} I_{rms} \cos \phi$$

$$\text{Power factor, } \cos \phi = \frac{\text{real power}}{\text{virtual power}}$$

$$\cos \phi = \frac{P_{av}}{E_{rms} I_{rms}} \quad \cos \phi = \frac{R}{Z}$$

(i) R only $\phi = 0$, $\cos \phi = 1$

$$P = I_{rms}^2 R = E_{rms}^2 / R$$

(ii) C only $\phi = 90^\circ = -\pi / 2$,

$$\cos \phi = 0 \quad P_{av} = 0$$

Series LCR

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right) \quad \cos \phi = \frac{R}{Z}$$

$$P_{av} = I_{rms}^2 R = \frac{E_{rms}^2 R}{Z^2}$$

(vii) At resonance

$$\phi = 0, \cos \phi = 1, \quad P_{av} = I_{rms}^2 R = \frac{E_{rms}^2}{R}$$

(ix) Parallel LCR

$$\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(\frac{1}{X_L} - \frac{1}{X_C} \right)^2}$$

86. Maxwell's Equations

$$(i) \oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} \quad (\text{Gauss's law of electricity})$$

$$(ii) \oint \vec{B} \cdot d\vec{s} = 0 \quad (\text{Gauss's law of magnetism})$$

$$(iii) \oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt} \quad (\text{Faraday's law})$$

87. (i) Energy of photon

$$E = h\nu, \quad E = hc / \lambda$$

$$(ii) \text{Momentum } p = \frac{E}{c}; p = \frac{h}{\lambda}$$

(iii) Rest mass = 0

$$(iv) \text{Mass equivalent to energy } m = \frac{E}{c^2}$$

88. Law of Reflection

$$\angle i = \angle r$$

Image Formed Between Two Inclined Plane Mirrors

$$(i) n = \frac{360^\circ}{\theta} - 1 \quad (\text{when } \frac{360^\circ}{\theta} \text{ is an even number})$$

$$(ii) n = \frac{360^\circ}{\theta} \quad (\text{when } \frac{360^\circ}{\theta} \text{ is an odd number})$$

NOTE: For symmetrical positions

$$n = \frac{360^\circ}{\theta} - 1$$

$$\text{Focal Length } f = \frac{R}{2}$$

where R = radius of curvature.

$$\text{Mirror Formula, } \frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

where u = object distance from pole,
v = image distance from pole.

89. Linear magnification

$$m = \frac{I}{O}$$

I = size of image, O = size of object.

90. Law of Refraction

$$\text{Snell's law : } n = \frac{\sin i}{\sin r}$$

Other Relations

$$n = \frac{v_1}{v_2}, \quad n = \frac{c}{v}$$

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{air}}}{n} \quad V_{\text{medium}} = \frac{V_{\text{air}}}{n} \quad n_1 \sin i = n_2 \sin r$$

(Symbol for refractive index changed from n to μ for practice).

$${}_1\mu_2 = \frac{\sin i}{\sin r}, \quad {}_1\mu_2 = \frac{1}{{}_2\mu_1}$$

$$(i) \text{ Lateral shift} = \frac{t \sin(i-r)}{\cos r}$$

$$(ii) \text{ Apparent depth} = \frac{\text{Real depth}}{\mu}$$

$$\text{Raised by } x = t \left(1 - \frac{1}{\mu} \right)$$

(iii) Total internal reflection

Critical angle = i_c

$$\sin i_c = 1 / \mu$$

For a luminous body at d depth radius of bright circular path at surface

$$r = d \tan i_c, \quad r = \frac{d}{\sqrt{\mu^2 - 1}}$$

Prism

$$i + e = A + \delta \quad \& \quad r_1 + r_2 = A$$

(i) At minimum deviation

$$i = e$$

$$r_1 = r_2$$

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin(A/2)}$$

(ii) For small angle prism

$$\delta_m = (\mu - 1)A$$

Dispersion

$$(i) \text{ Cauchy's relation } \mu = A + \frac{B}{\lambda^2}$$

$$\mu_{\text{red}} < \mu_{\text{violet}} \quad \delta_{\text{red}} < \delta_{\text{violet}}$$

(ii) Angular dispersion $\theta = \delta_V - \delta_R$

$$\theta = (\mu_v - \mu_R) A$$

(iii) Dispersive power

$$\omega = \frac{\delta_V - \delta_R}{\delta_Y} \text{ or, } \omega = \frac{\mu_V - \mu_R}{\mu_Y - 1} \text{ or, } \omega = \frac{\mu_B - \mu_R}{\mu_Y - 1}$$

Spherical Refracting Surface

(i) When object is situated in the rarer medium

$$-\frac{\mu_1}{u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

where, μ_1 = refractive index of rarer medium,

μ_2 = refractive index of spherical refracting surface.

(ii) When object is situated in denser medium

$$-\frac{\mu_2}{u} + \frac{\mu_1}{v} = \frac{\mu_1 - \mu_2}{R}$$

(iii) Power of spherical refracting surface

$$P = \frac{\mu_2 - \mu_1}{R}$$

Lens Maker's Formula,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

where f = focal length of lens, μ = refractive index of lens, R_1 and R_2 = radii of curvature of two refracting surfaces of a thin lens.

$$\text{Lens Formula, } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Power of Lens

$$P = \frac{1}{f(\text{in meters})}$$

Two Lenses Placed in Contact

$$(i) \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

where f_1 and f_2 = focal length of two lenses

f = focal length of the combination

$$(ii) \text{ Power of combination } P = P_1 + P_2$$

$$(iii) \text{ Magnification of combination, } m = m_1 \times m_2$$

Magnifying Power of Simple Microscope,

$$M = \left(1 + \frac{D}{f}\right) \text{ when final image is formed at } D$$

where D = least distance of distinct vision.

$$M = \left(\frac{D}{f}\right) \text{ when image is formed at infinity}$$

Magnifying Power of Compound Microscope,

$$M = -\frac{v_0}{u_0} \left(1 + \frac{D}{f_e}\right)$$

where v_0 = distance of image from object lens,

u_0 = distance of object from object lens,

f_e = focal length of eye lens.

Astronomical Telescope

(i) Magnifying power in normal adjustment,

$$M = -\frac{f_0}{f_e}$$

(ii) When the final image is formed at least distance of distinct vision, magnifying power

$$M = -\frac{f_0}{f_e} \left(1 + \frac{f_e}{D}\right)$$

91. In a compound microscope, the intermediate image is real inverted and magnified.

92. A lens behaves as a converging lens in air and a diverging lens in water. The refractive index of the material is between unity and 1.33.

WAVE OPTICS

Interference

Superposition of Waves of Equal Frequency and Constant Phase Difference

(i) Resultant wave amplitude

$$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$$

(ii) Resultant wave intensity

$$I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi$$

If $a_1 = a_2 = a_0$, and $I_1 = I_2 = I_0$, then

$$a = 2a_0 \cos (\phi / 2) \quad I = 4I_0 \cos^2 (\phi / 2)$$

Constructive Interference

Condition $\phi = 2n\pi \equiv 0, 2\pi, 4\pi, 6\pi, \dots \pi$

$$\Delta x = n\lambda \equiv 0, \lambda, 2\lambda, \dots \quad a_{\max} = a_1 + a_2$$

$$I_{\max} \propto (a_1 + a_2)^2 \quad I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{\max} = 4I_0, \text{ If } I_1 = I_2 = I_0$$

Destructive Interference

Conditions $\phi = (2n - 1)\pi = \pi, 3\pi, 5\pi, \dots$

$$\Delta x = (2n - 1)\frac{\lambda}{2} = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots$$

$$a_{\min} = a_1 - a_2 \quad I_{\max} \propto (a_1 - a_2)^2$$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2 \quad I_{\min} = 0, \text{ If } I_1 = I_2 = I_0$$

Young's Double Slit Experiment

(i) Position of n^{th} fringe on the screen

$$(a) \text{ Bright fringe } Y_n = \frac{n\lambda D}{d}$$

$$(b) \text{ dark fringe } Y_n = \frac{(2n - 1)\lambda D}{2d}$$

(ii) Fringe width

$$(a) \text{ linear } \beta = \frac{\lambda D}{d} \quad (b) \text{ angular } \alpha = \frac{\lambda}{d}$$

$$\beta_{\text{liquid}} = \frac{\beta_{\text{air}}}{\mu_{\text{liquid}}}$$

(iii) Shift of the central fringe by thin sheet

$$\text{Shift} = \frac{(\mu - 1)tD}{d}$$

Thin Films For reflected light

$$2\mu t \cos r = n\lambda \quad (\text{dark fringe})$$

$$2\mu t \cos r = \left(n - \frac{1}{2}\right)\lambda \quad (\text{bright fringe})$$

Single slit experiment

$$a \sin \theta = \Delta x, \Delta x = n\lambda \text{ for minima}$$

$$\Delta x = (2n+1)\lambda/2 \text{ for maxima}$$

(i) Telescope

$$\text{Limit of resolution } \theta = \frac{1.22\lambda}{a}$$

$$\text{Resolving power} = \frac{a}{1.22\lambda}$$

(ii) Microscope

$$\text{Limit of resolution} = \frac{1.22\lambda}{2\mu \sin \theta} = \frac{0.61\lambda}{\mu \sin \theta}$$

$$\text{Resolving power} = \frac{\mu \sin \theta}{0.61\lambda}$$

$$\text{Numerical aperture} = \mu \sin \theta$$

(iii) Eye Limit of angle of resolving

$$= 1 \text{ min of arc} = \frac{1}{60} \text{ degree.}$$

93. Polarized glass is used in sun glasses because it reduces the light intensity to half an account of polarization.

94. The number of photo electrons emitted per second from a metal surface increases when the intensity of the incident light increases.

Dual Nature of Matter and Radiation

95. Photoelectric Effect

(i) Work function $W = h\nu_0$

where h = Planck's constant

ν_0 = threshold frequency

(ii) Einstein's equation of photoelectric effect

(a) Maximum kinetic energy of the emitted photoelectrons

$$\frac{1}{2}mv^2 = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

where m = mass of the ejected photoelectron.

λ = wavelength of the incident radiation

λ_0 = threshold wavelength.

$$(b) \frac{1}{2}mv_{\max}^2 = h\nu - h\nu_0$$

where, $v_{\max}$ = maximum velocity of photoelectron,

ν_0 = frequency of incident radiation,

ν = threshold frequency.

96. de-Broglie Waves

$$\text{Wavelength } \lambda = \frac{h}{\sqrt{2mKE}}$$

97. For Photons

$$E = h\nu, E = \frac{hc}{\lambda}, p = \frac{h\nu}{c}, p = \frac{E}{c}$$

$$m = \frac{h\nu}{c^2}, m = \frac{h}{c\lambda} \quad \text{rest mass} = 0$$

$$\text{charge} = 0, \text{ spin} = 1 \left(\frac{h}{2\pi} \right)$$

Number of photons of wavelength λ emitted in t second from a lamp of power p ,

$$n = \frac{Pt\lambda}{hc}$$

98. Matter Waves

$$\lambda = \frac{h}{p}, \lambda = \frac{h}{mv}, \lambda = \frac{h}{\sqrt{2mE}},$$

$$p = \sqrt{2mE}, E = \frac{1}{2}mv^2 = eV$$

$$(i) \text{ For electron, } \lambda_e = \frac{12.27 \text{ \AA}}{\sqrt{V}}$$

(ii) For proton, $\lambda_p = \frac{0.286 \text{ \AA}}{\sqrt{V}}$

(iii) For alpha, $\lambda_\alpha = \frac{0.101 \text{ \AA}}{\sqrt{V}}$

(iv) For particle at temperature T (kelvin)

$$\lambda = \frac{h}{\sqrt{3mkT}} \text{ is } \left(E = \frac{3}{2} kT \right)$$

(v) For neutron or proton

$$\lambda = \frac{25.2 \text{ \AA}}{\sqrt{T}} \text{ (if } E = 3/2 kT, \text{ average energy)}$$

$$\lambda = \frac{30.8 \text{ \AA}}{\sqrt{T}} \text{ (if } E = kT, \text{ most probable energy)}$$

Accelerating potential to obtain de-Broglie wavelength λ_e

$$V = \frac{150.6}{\lambda_e^2} \text{ volt}$$

Condition for stable orbit $2\pi r_n = n\lambda$

99. Davisson-Germer Experiment

$$2d \sin \phi = n\lambda$$

$$D \sin \phi = n\lambda$$

100. Useful ratios,

$$\frac{\lambda_2}{\lambda_1} = \frac{m_1 v_1}{m_2 v_2}$$

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{m_1 E_1}{m_2 E_2}}$$

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{m_1 q_1 V_1}{m_2 q_2 V_2}}$$

$$\frac{\lambda_{ph}}{\lambda_e} = \sqrt{\frac{2m_e c^2}{E}}$$

101. Rutherford α -Scattering exp.

$$N(\theta) \propto \text{cosec}^4(\theta/2),$$

$$\text{Impact parameter } b = \frac{(Ze^2) \cot(\theta/2)}{(4\pi\epsilon_0)E}$$

(i) Distance of closest approach

$$r_0 = \frac{2Ze^2}{(4\pi\epsilon_0 E)}$$

102. Bohr Model

$$L = mvr = \frac{nh}{2\pi}$$

$$h\nu = E_i - E_f$$

(i) Radius of n^{th} orbit

$$r_n \propto \frac{n^2}{Z}$$

$$r_n = \frac{n^2}{Z} \frac{h^2}{4\pi^2 m k e^2}$$

(ii) Bohr radius

$$a_0 = \frac{h^2}{4\pi^2 m k e^2} = 0.529 \text{ \AA}$$

(iii) Ratio of radii

$$r_1 : r_2 : r_3 = 1 : 4 : 9$$

(iv) Velocity of electron in n^{th} orbit

$$v_n = \frac{Z}{n} \cdot \frac{c}{137}$$

$$v_n = \frac{Z}{n} \cdot \alpha c,$$

$$\alpha = 2\pi k e^2 / hc$$

$$\alpha = 1/137 \text{ (fine structure constant)}$$

$$v_1 : v_2 : v_3 : \dots = 1 : \frac{1}{2} : \frac{1}{3} : \dots$$

(v) Energy

$$U = \frac{kZe^2}{r}$$

$$K = \frac{1}{2} m v^2$$

$$K = -\frac{1}{2} U; E = \frac{1}{2} U$$

$$E = -K$$

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV} = -\frac{Z^2}{n^2} \left(\frac{2\pi^2 m k^2 c^4}{h^2} \right)$$

$$E_n = -\frac{2.18 \times 10^{-18} Z^2}{n^2} J$$

(vi) Ionization energy

$$= E_1 = (13.6 Z^2) eV$$

H-atom = 13.6 eV

He⁺ ion = 54.4 eV

Li⁺⁺ ion = 122.4 eV

(vii) Ionization potential

H-atom = 13.6 V

He⁺ ion = 54.4 V

(viii) Series formula (wave number $\bar{\nu} = 1/\lambda$)

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$R = 2\pi^2 mk^2 e^4 / ch^3$$

$$R = 1.097 \times 10^7 \text{ m}^{-1}$$

(ix) Series formula for H-atom

Lyman

$$\frac{1}{\lambda} = R \left(1 - \frac{1}{n^2} \right), n = 2, 3, 4, \dots$$

Balmer

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right), n = 3, 4, 5, \dots$$

Paschen

$$\frac{1}{\lambda} = R \left(\frac{1}{3^2} - \frac{1}{n^2} \right), n = 4, 5, 6, \dots$$

Brackett

$$\frac{1}{\lambda} = R \left(\frac{1}{4^2} - \frac{1}{n^2} \right), n = 5, 6, 7, \dots$$

103. Pfund

$$\frac{1}{\lambda} = R \left(\frac{1}{5^2} - \frac{1}{n^2} \right); n = 6, 7, 8, \dots$$

(i) Series limits ($\lambda_{\text{shortest}}$)

Lyman $\lambda_s = 912 \text{ \AA}$

Balmer $\lambda_s = 3645 \text{ \AA}$

Paschen $\lambda_s = 8201 \text{ \AA}$

(ii) Number of emission lines from excited state n

$$= n(n-1)/2$$

(iii) Time period of revolution

$$T_n \propto n^3 / Z^2$$

$$T_1 = 1.5 \times 10^{-16} \text{ s}$$

$$T_1 : T_2 : T_3 = 1 : 8 : 27$$

(iv) Frequency of revolution

$$\nu_n \propto Z^2 / n^3$$

$$\nu_1 = 6.6 \times 10^{15} \text{ Hz}$$

$$\nu_1 : \nu_2 : \nu_3 = 1 : \frac{1}{8} : \frac{1}{27}$$

(v) Current due to orbital motion

$$I_n \propto Z^2 / n^3$$

$$I_1 = 1 \text{ mA}$$

104. Disintegration Law

$$\frac{dN}{dt} = -\lambda N$$

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = \left(\frac{1}{2} \right)^{t/T}$$

105 Half-life and Decay Constant

$$\lambda = -\frac{dN/dt}{N}$$

$$\lambda N = \log_e 2$$

$$T = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T}$$

106. Mean Life

$$\tau = \frac{1}{\lambda}, \lambda = \frac{1}{\tau}$$

$$T = 0.693\tau, \tau = 1.443T$$

107. Activity

$$R = \left| \frac{dN}{dt} \right|$$

$$R = \lambda N$$

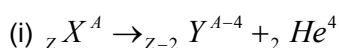
$$R = R_0 e^{-\lambda t}$$

$$\frac{R}{R_0} = \left(\frac{1}{2} \right)^{t/T}$$

$$1 \text{ Bq} = 1 \text{ decay/s}$$

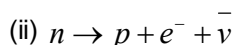
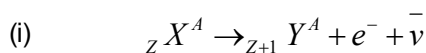
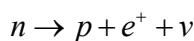
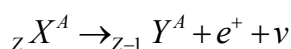
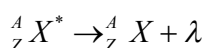
$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ dis/s}$$

$$1 \text{ Ci} = 37 \text{ GBq}$$

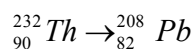
108. Alpha Decay


$$(ii) Q = (M_X - M_Y - M_\alpha)$$

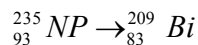
$$(iii) \text{Range } R_\alpha \propto E_\alpha^{3/2}$$

109. Beta Decay

110. Positron emission

111. Gamma Decay

112. Radioactive Series

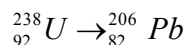
(i) $4n$ Thorium



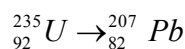
(ii) $4n + 1$ Neptunium



(iii) $4n + 2$ Uranium



(iv) $4n + 3$ Actinium


113. Radioactive Equilibrium

$$N_A \lambda_A = N_B \lambda_B$$

114. Decay Constant for Two Channels

$$(i) \lambda = \lambda_1 + \lambda_2$$

$$(ii) \frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$$

$$(iii) T = \frac{T_1 T_2}{T_1 + T_2}$$

115. Gamma Intensity Absorption

$$I = I_0 e^{-\mu x}$$

(i) Half thickness

$$x_{1/2} = \frac{0.693}{\mu}$$

(ii) Atomic mass unit

$$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$$

$$1 \text{ amu} \equiv 1 \text{ u}$$

$$1 \text{ u} \equiv 931.5 \text{ MeV}$$

(iv) Radius

$$R = R_0 A^{1/3}$$

(v) Volume

$$V \propto A$$

$$V = \frac{4\pi R_0^3}{3} A$$

(vi) Density

$$\rho = 2.4 \times 10^{17} \text{ kg m}^{-3}$$

= independent of A.

116. Mass Defect

$$\Delta M = Zm_p + (A - Z)m_n - M$$

Mass Defect

$$\Delta = -\Delta M$$

117. Packing Fraction

$$Pf = \frac{\Delta M}{A} = \text{mass defect per nucleon.}$$

118. Binding Energy

$$BE = (\Delta M)c^2$$

119. Binding Energy per Nucleon

$$\overline{BE} = \frac{BE}{A}$$

120. Uncertainty Principle

$$\Delta x \Delta p_x \geq (h / 4\pi),$$

$$\Delta E \Delta t \geq (h / 4\pi)$$

$$\Delta E = \frac{h}{4\pi\Delta t}$$

121. The amplification produced by a triode is due to the action of GRID.

122. Canal rays consist of Positive ions.

123. The curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for helium nucleus is very stable.

Electronic Devices

124. Band Gap

Metals : no gap

Insulators : $E_g \sim 6eV$

Semiconductors : $E_g \sim 1eV$

Si (Z=14), $E_g = 1.12eV$

Ge (Z=32) $E_g = 0.7eV$

125. Conductivity

(i) Metals : Very high $\sim 10^7$ to 10^8 S/m

(ii) Insulators : Very low $\sim 10^{-9}$ to 10^{-15} S/m

(iii) Semiconductors : intermediate

Si : $\sigma \sim 4.7 \times 10^{-4} Sm^{-1}$

Ge : $\sigma \sim 2.13 Sm^{-1}$

(resistivity is inverse of conductivity)

Metals : Conductivity decreases with temperature,

Semiconductors : Conductivity increases with temperature.

126. Temperature Coefficient of Resistance

Metal : positive

Semiconductor : negative

127. Conductivity of Semiconductors

(i) Intrinsic : $\sigma = e(n_e\mu_e + n_h\mu_h)$

$$\sigma = \sigma_0 e^{-E_g/2kT}$$

(ii) Extrinsic : n - type, $\sigma = en_e\mu_e$

p - type, $\sigma = en_h\mu_h$

128. p -type : Impurity III group, B, Al, Ga, In, Tl.

Majority charge carriers : holes

Minority charge carriers : electrons

129. n -type

(i) Impurity V group, P, As, Sb

(ii) Majority charge carriers : electrons

(iii) Minority charge carriers : hole

130. pn Junction

(i) Barrier potential ~ 0.5 V

(0.7 V for Si and 0.3 V Ge)

(ii) Width of depletion layer $\sim 1 \mu m \sim 10^{-6}$ m

(iii) Barrier electric field $E_B \sim 10^5$ to 10^6 V/m.

131. pn Junction Diode

(i) Forward resistance $r_f \sim \text{min}$

(ii) Reverse resistance $r_r \sim \text{very high}$ ($10^6 \Omega$)

$$I = I_0 (e^{eV/kT} - 1)$$

$$r_r = \Delta V / \Delta I, \text{ etc.}$$

(iii) Diode as rectifier formula (same as for vacuum tube)

132. Transistor

$$I_E = I_C + I_B$$

$$I_B \ll I_E,$$

$$I_B \ll I_C$$

$$I_C \leq I_E$$

133. Current Gain [In Common Base (CB)]

$$\alpha = \frac{I_C}{I_E},$$

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}$$

$$\beta = \frac{I_C}{I_B},$$

Current Gain [In Common emitter (CE)]

$$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}$$

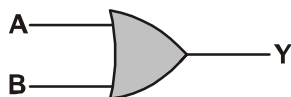
$$\alpha = \frac{\beta}{1 + \beta},$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

134. Logic Gates

(i) OR gate

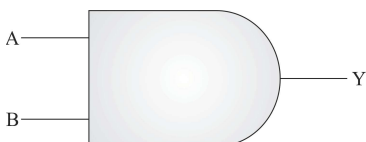
The OR gate operation is represented by the sign (+). If A and B are the two inputs and Y is the output, then the OR operation is represented as



logic symbol : $Y = A + B$ and is read as A OR B. It also termed as logical addition.

(ii) AND gate

The AND gate operation is represented by the dot (.) between the two variables or inputs



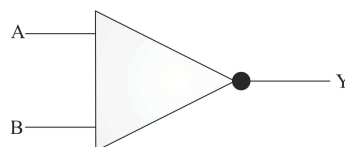
logic symbol

$$Y = A \cdot B$$

and is read as A AND B. The AND operation represents logical multiplication.

(iii) NOT gate

The NOT gate operation inverts the input or complements the input. It is represented by a bar (-) over the variable or the input.



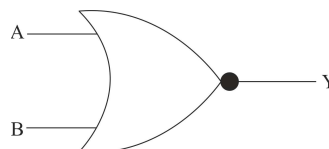
logic symbol

$$Y = \overline{A}$$

and is read as NOT A

(iv) NOR gate

The NOR gate is a combination of OR gate with a NOT gate. The Boolean expression for NOR gate is

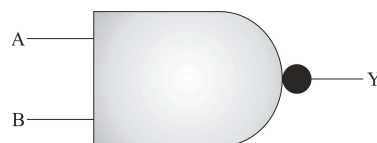


logic symbol

$$Y = \overline{A + B}$$

(v) NAND gate

If the output of the AND gate is connected to the input of a NOT gate, NAND gate is formed. The Boolean expression of NAND gate is



logic symbol

$$Y = \overline{A \cdot B}$$

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